06 / INDEPENDENT ANSWER KEY

Check the stock numbers.

Safi Services & Supplies | Fictional 30-day example
All monetary values are KES. The CSV is included for spreadsheet use.

Revenue	Cost of goods sold	Gross profit
179,640	149,345	30,295

Input data

Product	Opening	Sold	Unit cost	Price
Multipurpose cleaner 2L	120	95	310	380
Floor cleaner 5L	80	70	720	850
Hand soap 1L	150	40	260	300
Laundry soap bar	200	35	45	60
Bleach 1L	100	88	170	210
Microfibre pack	60	12	350	480
Toilet paper 10 pack	90	82	380	450
Bin liner box	40	8	900	1100

The calculation rules

Revenue = units sold × selling price
COGS = units sold × unit cost
Gross profit = units sold × (selling price - unit cost)
Closing stock = opening stock - units sold
Sell-through = units sold ÷ opening stock × 100
Top-up = max(0, units sold - closing stock)

Check two rows yourself

Floor cleaner: 70 × (850 - 720) = KES 9,100.
Multipurpose cleaner: 95 × (380 - 310) = KES 6,650.

06 / Independent answer key - continued

Profit, stock & assumptions

Profit contribution

Product	Revenue	COGS	Gross profit
Multipurpose cleaner 2L	36,100	29,450	6,650
Floor cleaner 5L	59,500	50,400	9,100
Hand soap 1L	12,000	10,400	1,600
Laundry soap bar	2,100	1,575	525
Bleach 1L	18,480	14,960	3,520
Microfibre pack	5,760	4,200	1,560
Toilet paper 10 pack	36,900	31,160	5,740
Bin liner box	8,800	7,200	1,600

Closing stock and simple top-ups

Product	Closing	Sell-through	Top-up
Multipurpose cleaner 2L	25	79.17%	70
Floor cleaner 5L	10	87.5%	60
Hand soap 1L	110	26.67%	0
Laundry soap bar	165	17.5%	0
Bleach 1L	12	88.0%	76
Microfibre pack	48	20.0%	0
Toilet paper 10 pack	8	91.11%	74
Bin liner box	32	20.0%	0

CLASSROOM ASSUMPTIONS

No restocking, returns, shrinkage, price changes or stockouts. Next-month demand is unchanged; delivery is instant and safety stock is zero. Real decisions need lead times, cash, demand, storage and accurate stock records. Overheads are absent: net profit is unknown.